Testing Accessibility Knowledge Across Pythia Model Sizes

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Introduction

Digital accessibility guidelines represent a narrow, specialized domain in web-scale training data. Terms like WCAG, ARIA, and alt text appear far less frequently than general web vocabulary, and their correct usage requires understanding both syntax and intent. This makes accessibility an unusual test case for studying emergence: the concepts are concrete enough to evaluate, rare enough to show scale sensitivity, and have clear right and wrong answers.

Most emergence research uses broad capability benchmarks — arithmetic, chain-of-thought reasoning, multilingual translation. These are useful but noisy. Accessibility concepts offer something different: a small, well-defined vocabulary with unambiguous ground truth and direct relevance to real-world tooling decisions.

This paper uses the Pythia model suite (160M–6.9B parameters) and TransformerLens to examine accessibility knowledge across four dimensions: whether models can define accessibility concepts, whether they can identify violations in code, whether recognition precedes generation, and how the 2.8B model internally represents a compound accessibility term. Pythia's controlled architecture — identical training data across all sizes — isolates scale as the variable of interest.

The findings are specific: accessibility knowledge emerges at 2.8B parameters for core concepts, later or not at all for acronyms, and evaluative capability lags behind declarative knowledge even at maximum scale tested. Attention pattern analysis shows mechanistic encoding of compound terms before full behavioral competence appears.

Related Work

Emergence in language models. Wei et al. (2022) established that language models exhibit emergent abilities — capabilities that appear sharply at specific parameter scales rather than improving gradually. This paper treats accessibility concept acquisition as a domain-specific case

study of emergence, using the Pythia suite's controlled architecture to isolate scale as the variable of interest.

Knowledge representation in transformers. Geva et al. (2021) demonstrated that feed-forward layers in transformers function as key-value memories, storing factual associations that attention heads retrieve and route. Subsequent work by Yao et al. (2024) mapped knowledge circuits in GPT-2 Medium using factual recall tasks, noting explicitly that their findings "may not be universally applicable to all knowledge domains." This paper examines whether domain-specific, low-frequency knowledge follows the same patterns.

Attention as an interpretability tool. Clark et al. (2019) showed that attention patterns in BERT encode linguistically meaningful relationships, establishing attention visualization as a valid method for probing internal representations. This work applies the same approach to compound accessibility terms, asking whether models bind multi-token concepts as units.

AI and digital accessibility. A growing body of work applies AI to accessibility tasks — generating alt text, detecting contrast failures, automating WCAG audits. This paper takes the inverse approach: rather than using AI as an accessibility tool, it examines what AI systems actually know about accessibility and when that knowledge emerges.

Methodology

Models

This study uses the Pythia model suite (Biderman et al., 2023): 160M, 410M, 1B, 2.8B, and 6.9B parameter models. Pythia models share identical architecture and training data across all sizes, isolating scale as the variable of interest. All models were loaded using TransformerLens (Nanda et al., 2022).

One architectural exception: Pythia 1B uses a different configuration than the other models in the suite — 16 layers, 8 heads, d_model=2048, d_mlp=8192, with parallel attention and MLP. The other models use 12 heads. This difference is relevant when interpreting per-head binding counts in Experiment 4; comparisons across model sizes account for this asymmetry.

Experiments

Experiment 1: Declarative Knowledge Ten accessibility concept prompts were run against each model using greedy decoding (temperature=0, max_new_tokens=10):

```
model = HookedTransformer.from_pretrained("pythia-2.8b")
output = model.generate(prompt, max_new_tokens=10, temperature=0)
```

Prompts covered core accessibility terms (screen reader, alt text, skip link), foundational acronyms (WCAG, ARIA), and general accessibility concepts (focus indicator, keyboard navigation, color contrast, semantic HTML, captions). Responses were evaluated against known correct definitions.

Experiment 2: Evaluative Knowledge Five code prompts were run against each model (temperature=0, max_new_tokens=20), asking models to identify accessibility violations in HTML snippets. Prompts covered missing alt attributes, non-semantic interactive elements, empty links, unlabeled inputs, and ambiguous link text. Responses were evaluated against correct accessibility explanations.

Experiment 3: Recognition vs. Generation Perplexity was computed for a correct and incorrect definition of "screen reader" across all model sizes:

```
correct = "A screen reader is software that reads text aloud for blind users."
wrong = "A screen reader is a device for viewing screens."

def get_perplexity(model, text):
    tokens = model.to_tokens(text)
    logits = model(tokens)
    log_probs = torch.nn.functional.log_softmax(logits, dim=-1)
    token_log_probs = log_probs[0, :-1, :].gather(1, tokens[0, 1:].unsqueeze(1)).squeeze()
    return torch.exp(-token_log_probs.mean()).item()
```

Lower perplexity indicates the model finds the text more expected. Comparing correct vs. incorrect perplexity across model sizes reveals whether recognition of correct definitions precedes the ability to generate them.

Experiment 4: Attention Pattern Analysis Attention weights were extracted across all layers and heads for the prompt "A screen reader is" using TransformerLens's activation cache:

```
logits, cache = model.run_with_cache(prompt)
attention = cache["pattern", layer]  # shape: [heads, seq, seq]
```

Token indices were verified for each model prior to analysis:

```
tokens = model.to_str_tokens(prompt)
print(list(enumerate(tokens)))
```

Binding strength between "reader" and "screen" was measured as the attention weight from the "reader" token to the "screen" token. Heads with attention weight above 0.1 were recorded. Weights above 0.5 were considered strong binding; weights between 0.2 and 0.5 moderate; weights below 0.2 were treated as background noise and excluded from analysis. This experiment was run for Pythia 2.8B and 6.9B. Full attention binding data for all model sizes is available in the project repository.

Reproducibility

All experiments were run with temperature=0 for deterministic outputs. Models 160M through 2.8B were run locally on an Apple M5 MacBook Pro (24GB RAM). Pythia 6.9B was run on Google Colab with an A100 GPU. Full notebooks are available at https://github.com/trishasalas/mech-interp-research/blob/main/pythia/pythia-a11y-emergence.ipynb.

Results

Experiment 1: Declarative Knowledge

Models were tested on ten accessibility concept prompts across all five Pythia sizes. Results show a sharp threshold pattern for core accessibility terms, with acronyms behaving differently.

Correct = matches known definition; Partial = incomplete or imprecise; Incorrect = wrong, off-topic, or loops

Prompt	160M	410M	1B	2.8B	6.9B
A screen reader is	×	×	≈	✓	≈
WCAG stands for	×	×	×	×	✓

Prompt	160M	410M	1B	2.8B	6.9B
A skip link is	×	≈	×	✓	×
The purpose of alt text is	×	×	×	✓	✓
ARIA stands for	×	×	×	×	×
A focus indicator is	×	×	×	×	×
Keyboard navigation allows	×	×	×	×	×
Color contrast is important because	×	×	×	×	×
Semantic HTML helps	×	×	×	×	×
Captions are used for	×	≈	×	≈	×

Several patterns emerge from this data.

Threshold at 2.8B. Screen reader, skip link, and alt text all show correct or near-correct responses first at 2.8B. Below this threshold, responses are either wrong or incomplete.

Screen reader shows non-monotonic behavior. The 2.8B model produces "reads aloud the text," correctly capturing the auditory output purpose. The 6.9B model produces "reads text on a computer screen," dropping "aloud." The larger model is more verbose but less precise on the critical detail. This mirrors the skip link result, where 2.8B is correct and 6.9B regresses. Capability near emergence thresholds can be unstable.

WCAG emerges at 6.9B; ARIA never emerges. Both are foundational accessibility acronyms. WCAG requires 6.9B parameters to expand correctly. ARIA fails at every scale tested — at 160M it produces gibberish, at 410M it loops, and from 1B onward it produces confident but wrong expansions ("Artificial Replacement of a Human Being," "A Rational Approach to Information and Automation," "Association of Research Libraries in Africa"). Confidence increases with scale; accuracy does not. This suggests ARIA is rare enough in training data that even 6.9B parameters cannot reliably encode it, while increased scale produces more fluent confabulation rather than correct recall.

General accessibility concepts fail across all scales. Focus indicator, keyboard navigation, color contrast, and semantic HTML produce generic responses at every model size. These terms appear in web-scale training data but without accessibility-specific context. The models complete the prompts plausibly without encoding accessibility meaning.

Experiment 2: Evaluative Knowledge

Models were tested on five code prompts requiring identification of accessibility violations. This tests whether models can apply accessibility knowledge, not just produce definitions.

Correct = identifies the accessibility violation accurately; Partial = identifies some issue but not the core violation; Incorrect = wrong, off-topic, or loops

Prompt	160M	410M	2.8B	6.9B
`<img src='photo.jpg'>` missing what	×	×	×	×
`<div>` with onclick not accessible because	×	×	×	×
Problem with `<a href='#'></a>`	×	≈	×	×
`<input type='text'>` needs a	×	×	×	×
'Click here' button is bad because	×	×	✓	✓

There is a clear gap between declarative and evaluative knowledge. The 2.8B model correctly defines alt text but cannot identify that `<img src='photo.jpg'>` is missing one — it repeats the prompt and stalls at every scale tested. The question explicitly asks what is missing; no model answers "alt text."

The only evaluative success is the "Click here" prompt, which succeeds at 2.8B and 6.9B. Ambiguous link text may be more common in training data as a named anti-pattern than missing alt attributes.

The `<div>` onclick responses show a trajectory worth noting. Responses become more specific with scale — from loops at 160M to "not a form control" at 6.9B — without arriving at the correct explanation (keyboard inaccessibility, missing role and tabindex). The 6.9B answer is the closest, pointing toward interactivity expectations, but it does not identify the actual problem.

Experiment 3: Recognition vs. Generation

Perplexity measures how expected a sequence is to the model. Lower perplexity means the model finds the text more natural. Testing whether models assign lower perplexity to a correct definition than an incorrect one reveals whether recognition precedes generation.

```
correct = "A screen reader is software that reads text aloud for blind users."
```

```
wrong = "A screen reader is a device for viewing screens."
```

Model	Correct	Wrong	Ratio
160M	106.7	41.4	Wrong by 2.6x
410M	40.1	32.8	Wrong by 1.2x
1B	18.8	42.2	Correct by 2.2x
2.8B	13.6	54.9	Correct by 4.0x
6.9B	15.6	46.1	Correct by 3.0x

The preference flips between 410M and 1B. At 160M and 410M the model finds the wrong definition more natural; at 1B it prefers the correct one. This flip precedes the generation threshold by one model size — 1B recognizes the correct definition before 2.8B can produce it.

The 2.8B model shows stronger correct preference than 6.9B (4.0x vs 3.0x). This is consistent with the non-monotonic pattern observed in declarative results: the relationship between scale and capability is not strictly linear near emergence thresholds.

Experiment 4: Attention Pattern Analysis

Attention pattern analysis across all five Pythia models examines whether models treat “screen reader” as a compound concept or as two independent tokens, and how this binding pattern relates to behavioral emergence.

For each model, attention weights from “reader” to “screen” were extracted across all layers and heads. Heads with weight above 0.5 are considered strong binding; the full data is available in the project repository.

Model	Layers	Strong (0.5+)	Strong in L0-3	% Early	Last Strong Layer
160M	12	10	7	70%	11
410M	24	22	16	73%	9
1B	16	11	10	91%	6
2.8B	32	25	20	80%	29
6.9B	32	37	28	76%	30

Note: 1B's architectural difference (8 heads vs 12) affects raw head counts; see Methodology.

Several patterns emerge from this data.

Early layers dominate across all models. Between 70-91% of strong binding occurs in layers 0-3 regardless of model size. Compound term binding is established early in the forward pass at every scale tested.

Strong head count scales with model size. 160M has 10 strong binding heads; 6.9B has 37. 1B is the expected outlier given its different architecture.

Last strong layer tracks behavioral emergence. Below the 2.8B emergence threshold, strong binding drops off early — layer 11 for 160M, layer 9 for 410M, layer 6 for 1B. At 2.8B and above, strong binding persists deep into the network — layers 29 and 30 for 2.8B and 6.9B respectively. The models that cannot correctly define screen reader do not sustain compound binding through the network. The models that can, do.

2.8B is the inflection point. Total heads above threshold jumps from 28 (1B) to 101 (2.8B) — a 3.6x increase that coincides exactly with the behavioral emergence threshold identified in Experiment 1. This suggests that robust, sustained attention binding across many heads may be a mechanistic correlate of accessibility concept emergence, not merely a consequence of larger model size.

All six models show strong binding in layers 0-3, including 160M which cannot produce a correct definition. Whether early-layer binding at small scales reflects genuine compound representation or proximity effects cannot be determined from attention weights alone and is noted as a limitation.

Control Experiment: Ruling Out Proximity Effects To test whether the binding signal reflects compound concept encoding rather than simple token adjacency, a control experiment measured attention weights between non-compound token pairs at 2.8B. Two conditions were tested:

Adjacent function words: "She finished her homework, and then she watched a movie" — measuring attention from "then" (index 7) to "and" (index 6). This pair has high co-occurrence but no semantic dependency relationship.

Adjacent modifier-noun pair without disambiguation: "She drank a glass of cold water" — measuring attention from "water" (index 7) to "cold" (index 6). "Cold" modifies "water" but does not disambiguate it; "water" carries stable meaning without the modifier.

Both control pairs produced strong binding at L1H12 (0.95 and 0.94 respectively) when tokens were adjacent. However, when “then” was separated from “and” by one intervening token, L1H12 dropped out of the top results entirely, suggesting L1H12 is sensitive to token position rather than semantic relationship. Subsequent induction head testing confirmed that L1H12 exhibits previous-token head behavior — it attends systematically to the immediately preceding position regardless of content.

This characterization reframes the control comparison. L1H12’s consistent presence across all tested pairs, including accessibility compounds, reflects a general positional mechanism rather than compound-specific processing. The meaningful signal is not L1H12 specifically but the distribution across heads and layers. Accessibility compounds at 2.8B recruit 101-208 heads above threshold, with strong binding persisting to layers 29-30. Function word pairs recruit comparable head counts but do not show the same deep-network persistence pattern. The sustained late-layer binding — not the raw head count — is the distinguishing feature of compound semantic encoding.

A complete analysis isolating which heads are unique to accessibility compounds versus general adjacent-token binding is noted as a direction for future work.

Binding Generalizes Across Accessibility Compounds

Having ruled out proximity effects, attention binding was measured for two additional accessibility compound terms at 2.8B: alt text and skip link.

Compound	Total heads >0.1	Strong (0.5+)	Top score
screen reader	101	25	0.9909
alt text	211	49	0.9856
skip link	158	32	0.9816

All three compounds show the same pattern of early-layer concentration with deep-network persistence at 2.8B, with top binding scores above 0.98 in each case.

This rules out a compound-specific explanation. The binding pattern is not an artifact of how “screen reader” tokenizes or how frequently it appears in training data. It is a general property of accessibility compound terms at the 2.8B emergence threshold. The models that can define these concepts correctly show robust, distributed binding across many heads and many layers. The models that cannot show weak binding that drops off early.

Discussion

This paper examined accessibility concept acquisition across Pythia model scales using four complementary methods: behavioral generation, code evaluation, perplexity-based recognition, and attention pattern analysis. The results converge on a consistent picture with several implications for both accessibility tooling and emergence research.

The 2.8B Threshold Is Meaningful, Not Arbitrary

The behavioral findings show a sharp threshold at 2.8B parameters for core accessibility concepts. This is not a smooth gradient — 1B partially retrieves "reads text from a screen" while 2.8B produces "reads aloud the text," capturing the critical auditory output purpose. The threshold pattern, combined with the 3.6x jump in sustained attention binding heads at 2.8B, suggests a qualitative shift rather than incremental improvement. Below 2.8B, the model has partial encoding that cannot support reliable generation. At 2.8B, encoding is sufficient to sustain compound binding throughout the network and produce correct behavioral output.

Encoding and Retrieval Are Dissociable

The perplexity results demonstrate that models prefer correct definitions before they can produce them. At 1B, the model assigns lower perplexity to the correct screen reader definition despite generating "reads text from a screen" — missing the auditory purpose. This dissociation between recognition and generation suggests that accessibility knowledge accumulates in model representations before it becomes accessible through the generation pathway. The capability exists earlier than behavior indicates.

This has practical implications: perplexity-based probes may detect accessibility knowledge at smaller scales than behavioral tests suggest. Models that appear to fail accessibility benchmarks may have partial encoding that structured probing could surface.

The Declarative-Evaluative Gap Is Not Closed at Any Scale Tested

Even at 6.9B, models that correctly define accessibility concepts cannot reliably identify violations in code. The single evaluative success — identifying ambiguous link text — may reflect training data

frequency rather than genuine accessibility reasoning. The image missing alt attribute failure is particularly telling: a model asked explicitly what is missing produces the prompt back rather than "alt text," despite correctly defining alt text in declarative tests. Knowing and applying are different capabilities that do not emerge together.

For accessibility tooling practitioners, this gap is the central finding. A model that passes a definition benchmark is not ready for code review tasks. Evaluative capability requires either substantially larger scale than tested here or domain-specific training.

Acronym Failure Is Systematic

WCAG and ARIA show qualitatively different failure patterns than conceptual terms. Conceptual terms show emergence — partial at 1B, correct at 2.8B. Acronyms either emerge very late (WCAG at 6.9B) or not at all (ARIA at any scale tested). The ARIA failure is particularly informative: hallucinated expansions become more fluent and confident with scale without becoming accurate. At 160M, ARIA produces gibberish. At 6.9B, it produces a coherent but wrong expansion ("Association of Research Libraries in Africa") with apparent conviction.

This suggests WCAG and ARIA are rare enough in web-scale training data that standard scaling cannot reliably encode them. Specialized training data or fine-tuning is likely required for reliable acronym expansion in the accessibility domain.

Attention Binding as a Correlate of Emergence

The sustained attention binding pattern at 2.8B and 6.9B — strong binding persisting to layers 29-30 versus dropping off at layers 6-11 in smaller models — provides a mechanistic correlate for behavioral emergence. The models that can define screen reader sustain the compound binding across the full network. The models that cannot drop it early.

This finding is correlational, not causal. Whether sustained binding causes correct generation, or both are consequences of a third factor such as MLP encoding depth, cannot be determined from attention weights alone. The presence of strong early-layer binding in 160M — including one head at perfect score 1.0 — despite behavioral failure suggests early binding is necessary but not sufficient for emergence. Whether this reflects genuine compound representation or proximity effects in the absence of higher-level context remains an open question for future probing work.

Induction head testing confirmed that L1H12 exhibits previous-token head behavior — it attends systematically to the immediately preceding position regardless of content or semantic relationship. This is a known circuit component in transformer models and is distinct from compound-specific binding. The consistent appearance of L1H12 across accessibility compounds, function word pairs, and general modifier-noun pairs is explained by this positional mechanism rather than accessibility-specific encoding. The relevant mechanistic signal is the distribution and persistence of binding across the full head population, not the behavior of any individual head.

Relationship to Prior Work

Wei et al. (2022) established emergence as a general phenomenon. This paper treats accessibility concept acquisition as a domain-specific case study with properties that make it a productive probe: concepts are rare enough to show scale sensitivity, evaluable against clear ground truth, and directly relevant to real-world applications. The results confirm that emergence operates in specialized domains but with domain-specific timing — accessibility acronyms require substantially more scale than conceptual terms, and evaluative capability does not emerge within the range tested.

The finding that general accessibility concepts (focus indicator, keyboard navigation, color contrast, semantic HTML) fail across all scales suggests training data frequency is the limiting factor, not model capacity per se. These terms appear in web-scale data but without the accessibility-specific context required for meaningful representation.

Conclusion

This paper mapped accessibility concept acquisition across five Pythia model sizes using behavioral, perplexity-based, and mechanistic analysis. The findings establish a clear picture of how and when accessibility knowledge emerges in language models trained on web-scale data.

Core accessibility concepts — screen reader, alt text, skip link — emerge behaviorally at 2.8B parameters. Foundational acronyms require more scale (WCAG at 6.9B) or fail entirely (ARIA at all scales tested). General accessibility vocabulary fails across all model sizes, suggesting training data frequency as the limiting factor rather than model capacity. Evaluative capability — identifying accessibility violations in code — does not emerge within the range tested, even in models that correctly define the relevant concepts.

The perplexity results show that recognition precedes generation: models prefer correct definitions before they can produce them, with the preference flipping between 410M and 1B. The attention binding analysis shows that sustained binding of “screen reader” as a compound concept across the full network is a mechanistic correlate of emergence — present at 2.8B and 6.9B, absent at smaller scales.

For practitioners building accessibility tooling on language models, the central implication is that behavioral success on definition tasks does not predict success on evaluation tasks. The declarative-evaluative gap observed here suggests that reliable accessibility code review requires either substantially larger models or domain-specific training data that goes beyond the web-scale distribution.

For emergence researchers, accessibility concepts offer a useful probe domain: rare enough to show scale sensitivity, concrete enough to evaluate against ground truth, and distinct enough from general knowledge to isolate domain-specific acquisition. The ARIA confabulation pattern — increasingly fluent wrong answers with scale — illustrates a failure mode that definition-based benchmarks would not detect.

The mechanistic findings here are correlational. Whether sustained attention binding causes behavioral emergence or reflects a common underlying factor requires causal analysis beyond the scope of this paper, and is a direction for future work.

Limitations

This paper reports findings from a first investigation into accessibility concept emergence across Pythia model scales. Several limitations should be considered when interpreting the results.

Prompt Design

All experiments used completion-style prompts (“A screen reader is,” “WCAG stands for”) rather than instruction or question formats. Prompt framing is known to affect retrieval in language models, and this paper’s own findings support that sensitivity. The alt text retrieval experiment demonstrated that the HTML-framed prompt (“In HTML, the alt attribute provides a text description of an”) performed worse than the simpler framing despite testing identical knowledge. Results reported here reflect one prompt design choice and may not generalize to other framings.

The declarative prompts were adapted from attention binding experiments rather than designed independently for behavioral testing. Future work should compare multiple prompt formulations to establish which results are robust across framings.

Evaluative Experiment Prompts

The evaluative experiment used zero-shot code completion prompts. The image missing alt attribute result — where models loop the prompt rather than identify the missing attribute — may reflect document completion behavior rather than a knowledge failure. Models trained on code repositories may interpret `<img src='photo.jpg'>` as the beginning of a code listing and complete accordingly. Instruction-tuned models or explicit question formatting ("What accessibility attribute is missing from this HTML?") may produce different results. This limitation is noted but not resolved here.

Perplexity Experiment Scope

Recognition versus generation was tested using a single sentence pair for one concept (screen reader). The preference flip between 410M and 1B is a clear finding, but a single pair is insufficient to establish this as a general pattern. A broader set of correct/incorrect pairs across multiple accessibility concepts would strengthen the recognition-precedes-generation claim.

Attention Binding Methodology

The proximity control experiment was run at 2.8B only. Control testing revealed that adjacent token pairs — including function word pairs and modifier-noun pairs without disambiguation — produce strong binding at L1H12, which was subsequently confirmed as a previous-token head attending systematically to the immediately preceding position regardless of content. This finding limits the interpretation of L1H12's consistent appearance across accessibility compounds. Whether other heads show accessibility-specific binding patterns that do not appear in non-compound controls has not been fully characterized and is noted as a direction for future work. Early-layer binding at 160M (including Layer 11 Head 8 at score 1.0) remains ambiguous. Whether this reflects genuine compound representation or positional attention at small scales cannot be determined from attention weights alone.

The compound generalization finding — that alt text and skip link show comparable binding to screen reader at 2.8B — is based on head counts above threshold across all three compounds.

Hardware and Reproducibility

Models 160M through 2.8B were run locally on an Apple M5 MacBook Pro (24GB RAM) using MPS acceleration. The 6.9B model was run on Google Colab with an A100 GPU. While results were verified for consistency where possible, hardware differences introduce potential reproducibility variation. All notebooks are available in the project repository for independent verification.

Scale Coverage

The scale ladder covers 160M through 6.9B parameters. ARIA fails to emerge at any scale tested. Whether ARIA would emerge at larger scales is unknown. The evaluative gap — models that define concepts cannot identify violations — was not tested beyond 6.9B. Both findings may reflect limitations of the scale range rather than fundamental properties of these concepts.